THE VANISHING AND SURJECTIVITY STATEMENTS HOLD FOR THE EXTENSION BY ZERO OF A FINITE LOCALLY CONSTANT SHEAF OF F_ELL-VECTOR SPACES ALONG A CONNECTED OPEN OF A REDUCED CURVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the trace-method reduction, the Zariski main theorem factorisation with the identification V = U x_X Y, the shriek-pushforward compatibility, the transport along the finite morphism, and the final filtration step.

Editorial note. The label lemma-vanishing-easier names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing situation and to the enumerated statements by their labels; the eleven cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The trace method for a finite etale morphism of degree prime to ell. Same chapter, the section on the trace method and the lemma producing a finite etale morphism of degree prime to ell, giving maps G -> f_*f^{-1}G -> G composing to an isomorphism. Supplies the reduction replacing G by f_*f^{-1}G.
- Zariski's main theorem for a quasi-finite separated morphism. More on Morphisms, the lemma factoring a quasi-finite separated morphism through a finite one by an open immersion with dense image. Supplies the diagram with the finite fbar : Y -> X and the dense open immersion j'.
- Compatiblity of the shriek with a finite pushforward and the finite pushforward statements. Same chapter, the lemma equating the extension by zero of a finite pushforward with the finite pushforward of the extension by zero, and the lemma transporting the statements along a finite morphism. Supplies the reduction to j'_!f^{-1}G and then along the finite morphism.
- The existence of a filtration and the short exact sequence statements. Same chapter, the lemma giving a filtration of a finite locally constant sheaf by constant pieces, and the lemma on the behaviour of the statements in a short exact sequence. Supplies the final reduction to the constant sheaf Z/ellZ over a dense open of a reduced curve.

Source reference. Original reference 06AP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15365–15365. Prerequisite pointer invoked at proof

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line(s) 15611; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A53: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15335–15335. Prerequisite pointer invoked at proof line(s) 15611; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GJ1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11612–11612. Prerequisite pointer invoked at proof line(s) 15637; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A5D: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15471–15471. Prerequisite pointer invoked at proof line(s) 15641; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A3R: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10414–10414. Prerequisite pointer invoked at proof line(s) 15616; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A5C: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15450–15450. Prerequisite pointer invoked at proof line(s) 15646; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A3Q: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15566–15566. Prerequisite pointer invoked at proof line(s) 15649; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05K0: sources/stacks-chow/tags.txt, source lines 7184–7184. Prerequisite pointer invoked at proof line(s) 15625; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 03SH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10330–10330. Prerequisite pointer invoked at proof line(s) 15617; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A52: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15324–15324. Prerequisite pointer invoked at proof line(s) 15606; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0A52 assume X reduced. Let $j : U \to X$ an open immersion with U connected. Let ℓ be a prime number. Let $\mathcal{G}$ a finite locally constant sheaf of $\mathbf{F}_\ell$-vector spaces on U. Let $\mathcal{F} = j_!\mathcal{G}$. Then statements (0A53) – (06AP) hold for $\mathcal{F}$.*

Proof. Let $f : V \to U$ be a finite étale morphism of degree prime to ℓ as in Lemma 0A3R. The discussion in Section 03SH gives maps

$$\mathcal{G} \to f_*f^{-1}\mathcal{G} \to \mathcal{G}$$

whose composition is an isomorphism. Hence it suffices to prove the lemma with $\mathcal{F} = j_!f_*f^{-1}\mathcal{G}$. By Zariski's Main theorem (More on Morphisms, Lemma 05K0) we can choose a diagram

$$\begin{array}{ccc} V & \xrightarrow{j'} & Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \overline{f}} \\ U & \xrightarrow{j} & X \end{array}$$

with $\overline{f} : Y \to X$ finite and j' an open immersion with dense image. We may replace Y by its reduction (this does not change V as V is reduced being étale over U). Since f is finite and V dense in Y we have $V = U \times_X Y$. By Lemma 0GJ1 we have

$$j_!f_*f^{-1}\mathcal{G} = \overline{f}_*j'_!f^{-1}\mathcal{G}$$

By Lemma 0A5D it suffices to consider $j'_!f^{-1}\mathcal{G}$. The existence of the filtration given by Lemma 0A3R, the fact that $j'_!$ is exact, and Lemma 0A5C reduces us to the case $\mathcal{F} = j'_!\underline{\mathbf{Z}/\ell\mathbf{Z}}$ which is Lemma 0A3Q. □